

The Perceived Effects of Business Process Management

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Abstract—Several organizations choose to be process-oriented. They apply the concept of business process management and focus on business processes instead of emphasizing functional structures, i.e. they explicitly design and document their processes, implement the process owner role, employ process performance measurement, etc. This paper empirically explores the outcomes of business process management by conducting interviews with a total of 44 process-oriented firms. Key findings indicate that process orientation leads to higher transparency, clear responsibilities, higher efficiency, structure and tidiness, higher product quality, faster throughput times, and better customer orientation.

Process orientation, business process management, organizational performance

I. INTRODUCTION

Many organizations choose to be process-oriented. Process orientation (PO) means focusing on business processes ranging from customer to customer instead of placing emphasis on functional and hierarchical structures [1]. PO emphasizes process as opposed to hierarchies with special focus on outcomes, particularly customer satisfaction [2]. A process-oriented organization is also often referred to as “horizontal organization” [3], “process centered organization” [4], “process enterprise” [5], “process focused organization” [6] or simply “process organization” [7]. A process-oriented organization comprehensively applies the concept of business process management (BPM). A firm which adopted the process-view of its organization, regardless of whether it has already run through business process reengineering (BPR) and/or process improvement projects or not, is concerned with the management of its business processes [8]. Business process management does not only incorporate the discovery, design, deployment and execution of business processes, but also interaction, control, analysis and optimization of processes [9].

A lot of the existing literature of the process-oriented organization has been in the popular press and lacks research or an empirical focus [10]. The objective of this work is to empirically explore the effects of BPM. It does not examine the effects of BPR. The main difference between BPM and BPR is that the former refers to the ongoing management of an organization based on its business processes, whereas the latter is a single project concerned with the radical redesign of business processes.

The paper will begin by describing the PO construct, followed by a brief literature review on the effects of PO. Next, the research design of the study is specified. Finally the results are discussed followed by a conclusion of this work.

II. DEFINING THE PROCESS ORIENTATION CONSTRUCT

Process orientation is a construct consisting of several dimensions:

First, a prerequisite for managing an organization based on its processes is to know which business processes are performed within the organization and how they are related to each other [1], [11], [12], [13]. Hence, a process-oriented firm explicitly designs and documents its business processes.

Second, management needs to support the process program. Without the support of senior executives, the process idea cannot unfold its full potential [11], [12].

Third, the existence of process owners is the most visible difference between a process enterprise and a traditional organization [5]. A business process needs to have a manager who has end to end responsibility of the process.

Fourth, a process-oriented organization comprehensively applies the concept of process performance measurement. By focusing measurement on processes rather than functions, alignment and common focus across separate organizational units can be achieved [14]. Implementing measures and taking corrective actions are operating precepts of process management [15].

Finally, there are other characteristics a process-oriented organization exhibits, including a process-oriented corporate culture (e.g. teamwork [12], readiness to change [12], and customer focus [1], [12]), IT systems which seamlessly support business processes [1], [12], a process-oriented organizational structure [12], [16], people and expertise (e.g. existence of process redesign and change management experts [12]), process-oriented HR systems (e.g. existence of an incentive system emphasizing the process' needs [12], [17]) and the existence of a BPM office coordinating and integrating process projects [12], [18].

III. LITERATURE REVIEW ON THE EFFECTS OF PROCESS ORIENTATION

Many authors argue that PO yields to a range of benefits. Hammer [12] states that companies of all sizes and in virtually every industry have achieved extraordinary improvements in cost, quality, speed, profitability, and other key areas by applying BPM. Schmelzer and Sesselmann [19] argue that integrated business process management yields to higher customer satisfaction, lower cycle times, increase of process and product quality, reduction of costs, increase of sales, higher profits and higher company value. Hinterhuber [11] argues that process management yields to an improvement of product quality, service quality, customer satisfaction and productivity.

Frei, Kalakota, Leone and Marx [20] empirically explored the relation between branched-based processes and financial performance of U.S. retail banks. One of the findings of this study is that banks which perform better across those branch-based processes also tend to have a better financial performance. The empirical study of Gustafsson, Nilsson and Johnson [21] analyzed quality practices in Swedish service organizations. The study shows direct impact of process orientation on customer satisfaction for large service organizations. The results of the empirical study of McCormack and Johnson [13] give evidence that process orientation helps companies to improve business performance, reduce inter-functional conflict and improve “esprit de corps”. In this study, manufacturing firms across multiple industries in the United States were analyzed. Silvestro and Westley [22] carried out case study research to explore the operational changes resulting from reengineering companies along process. One large electronics company and one large retail company (both U.K. based organizations) were investigated. Perceived benefits were increased market responsiveness, improved collaboration between functions and alignment of organizational objectives. They also report disadvantages like duplication of functional expertise and increased operational complexity resulting in an escalation of cost. A case study of Volvo car company conducted by Hertz, Johansson and de Jager [23] reported that Volvo’s shift to a process organization resulted in lower inventory costs, shortened lead-times, increased delivery reliability, and higher customer satisfaction.

IV. RESEARCH DESIGN

The present study on the effects of PO is carried out as a part of on-going research investigating Austrian corporations operating in metal and machinery industry. Since the application of BPM becomes more important at a certain firm size, small organizations (i.e. firms with less than 50 employees) were excluded from the study. The population of this study is therefore defined as Austrian corporations operating in metal and machinery industry with at least 50 employees. Firms were selected randomly and telephone interviews were used for data collection. All telephone interviews were personally conducted by the researcher. For every firm one executive (CEO, CIO or quality manager) was interviewed. Data quality of this study is expected to be high since respondents were personally identified and interviewed assuring that the interviewee has the knowledge to truthfully answer the questions.

An interview began by asking several questions to assure if BPM is applied within the firm. Questions about several PO dimensions, such as design and documentation of business processes, management commitment towards the process approach, the process owner role, process performance measurement etc. were asked to get a picture of the firm’s BPM effort. If a firm turned out to be engaged in BPM (i.e. if the firm worked on at least two of the aforementioned PO dimensions), the respondent was further asked about the experienced benefits of PO. Basically, one completely open question about the perceived benefits was asked. The intention was not to guide the interviewees through a specific set of questions (i.e. questions about the perceived effects of PO on a predefined set of outcome variables, such as product quality, customer satisfaction or financial performance) but the objective was to let the interviewees report which effects of PO they profoundly had perceived and therefore had well-kept in their mind. A total of 152 organizations were contacted whereas 44 firms were willing to give an interview as well as turned out to be engaged in BPM and therefore remained in the sample.

V. RESULTS

The answers of the interviews were analyzed and classified, leading to the following results:

- PO leads to *better transparency*: 18 respondents (that is 40.91% of the total respondents) stated that by applying BPM, the organization and/or business processes became more transparent and understandable. This leads to better identification of organizational problems and their causes. In particular, by applying process performance measurement, arising problems can be identified and counter measures can be initiated very quickly.
- Nine respondents (that is 20.45%) reported that process-oriented organizational approach leads to *clear responsibilities* because of the process owner role terminating many unclarities caused by fragmented and/or blurry accountability.
- Eight respondents (that is 18.18%) refer to the gained *efficiency/productivity* due to PO. Non value-adding activities are better identified and can be called into question.
- Seven respondents (that is 15.91%) reported that PO brings *clear structure, tidiness, and clear organizational interfaces*.
- Six respondents (that is 13.64%) stated that a PO approach yields an *improvement of product quality*.
- There were also six respondents (that is 13.64%) who reported that PO yields *process speed improvements*.
- Six respondents (that is 13.64%) stated that PO leads to a better *customer orientation*, since the firm’s business processes are aligned with the needs of the internal and/or external customer.

- Five respondents (that is 11.36%) reported that PO *facilitates communication* within the organization which can be partly attributed to the transparency achieved through PO.
- Five respondents (that is 11.36%) reported that PO encourages cross-departmental thinking (it “bridges the functional silos”) and yields *better collaboration* between departments.
- Five respondents (that is 11.36%) reported that PO leads to *growth* or acts as an enabler for growth.
- Four respondents (that is 9.09%) reported that PO leads to an *improvement of financial performance*, in terms of e.g. profitability.
- Three respondents (that is 6.82%) stated that the PO approach helps to *deliver products on-time*.
- Three respondents (that is 6.82%) stated that the PO approach yields *higher customer satisfaction*.
- Two respondents (that is 4.55%) referred to the fact that PO leads to *higher employee motivation and satisfaction*.
- Two respondents (that is 4.55%) referred to the fact that PO leads to *better and faster initial job training*.
- Two respondents (that is 4.55%) stated that PO leads to an *improvement of process quality*, e.g. the errors within the process are minimized.
- One firm (that is 2.27%) reported that PO *facilitates serving a heterogeneous customer structure*.
- One firm (that is 2.27%) reported that PO has helped to improve the *ability to deliver* (which is the extent to which the organization is able to agree upon delivery times requested by customers).
- Finally, one firm (that is 2.27%) reported that PO led to *higher flexibility* in terms of being able to rapidly adapt the organization to changing environmental conditions.

The results are also shown in Figure 1.

VI. CONCLUSION

In this paper, an attempt has been made to empirically investigate the effects of process orientation and business process management. The effects most often reported include improvement of transparency, establishment of clear responsibilities, improvement of efficiency, establishment of structure and tidiness, and improvement of product quality, throughput times and customer orientation.

The present work on the effects of PO, carried out as a part of on-going research, comes with important research limitations. First, the sample in this work only included Austrian firms operating in metal and machinery industry. Thus, generalizability of the findings to other industries or other countries is open to scrutiny. Second, organizations were asked about the perceived benefits of BPM. Therefore, possible

disadvantages resulting from BPM are not reported by this study. Third, organizations were asked just one completely open question about the encountered benefits of BPM. Hence, only the top-of-mind gains were discussed, while other important but secondary benefits might not be noted.

There are some avenues for future research. First, this study could be applied to other industries and/or countries (of particular interest would be the service industry). Second, the performance effects could be assessed quantitatively by investigating the effects of PO on various outcome variables such as financial performance or non-financial performance. Finally, there is a need of studies exploring which individual PO constructs (e.g. design and documentation of business processes, the process owner role, process performance measurement, etc.) have an effect on organizational performance to which degree.

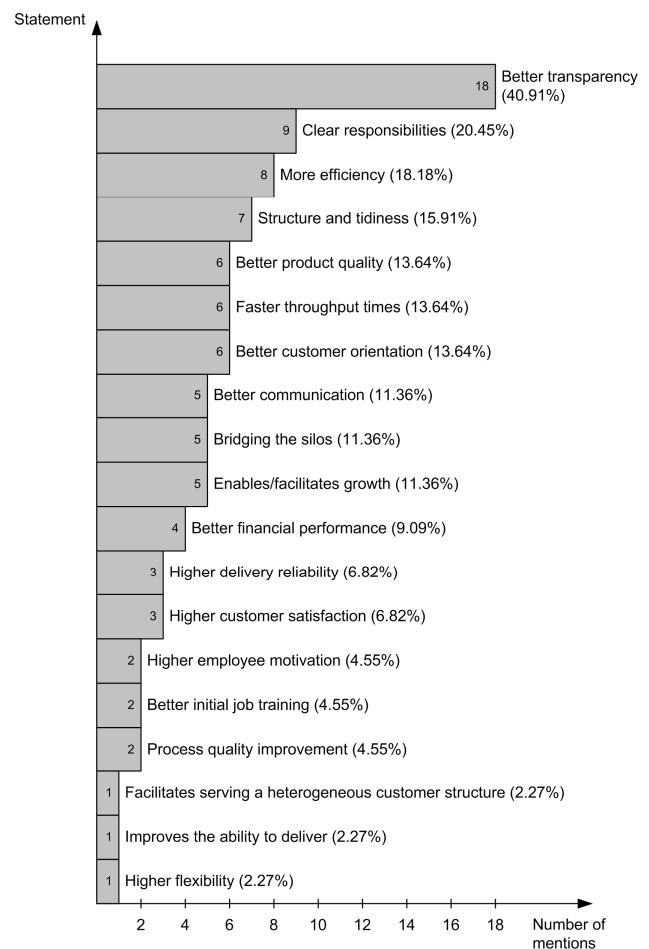


Figure 1. Perceived effects of process orientation. Answers of 44 interviewed firms are reported.

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REFERENCES

- [1] H. A. Reijers, "Implementing BPM systems: the role of process orientation," *Business Process Management Journal*, vol. 12, pp. 389–409, 2006.
- [2] K. McCormack and B. Johnson, "Business process orientation, supply chain management, and the e-corporation," *IIE Solutions*, pp. 33–37, October 2001.
- [3] F. Ostroff, *The Horizontal Organization*. New York, NY: Oxford University Press, 1999.
- [4] M. Hammer, *Beyond Reengineering*. London: Harper Collins, 1996.
- [5] M. Hammer and S. Stanton, "How process enterprises really work," *Harvard Business Review*, vol. 77, pp. 108–118, November-December 1999.
- [6] R. A. Gardner, *The Process-Focused Organization*. Milwaukee, WI: Quality Press, 2004.
- [7] M. Osterloh and J. Frost, *Prozessmanagement als Kernkompetenz*. Wiesbaden: Gabler, 2006.
- [8] C. Armistead and S. Machin, "Business process management: implications for productivity in multi-stage service networks," *International Journal of Service Industry Management*, vol. 9, pp. 323–336, 1998.
- [9] H. Smith and P. Fingar, *Business Process Management: The Third Wave*. Tampa, FL: Meghan-Kiffer Press, 2003.
- [10] A. P. Sussan and W. C. Johnson, "Strategic capabilities of business processes: looking for competitive advantage," *Competitiveness Review*, vol. 13, pp. 46–52, 2003.
- [11] H. H. Hinterhuber, "Business process management: the European approach," *Business Change & Re-engineering*, vol. 2, pp. 63–73, 1995.
- [12] M. Hammer, "The process audit," *Harvard Business Review*, vol. 85, pp. 111–123, April 2007.
- [13] K. P. McCormack and W. C. Johnson, *Business Process Orientation. Gaining the E-Business Competitive Advantage*. Boca Raton, FL: St. Lucie Press, 2001.
- [14] M. Hammer, "The 7 deadly sins of performance measurement," *MIT Sloan Management Review*, vol. 48, pp. 19–28, Spring 2007.
- [15] E. H. Melan, "Process management: a unifying framework for improvement," *National Productivity Review*, vol. 8, pp. 395–406, 1989.
- [16] B. Karlöf and F. Lövingsson, *The A-Z of Management Concepts and Models*. London: Thorogood Publishing, 2005.
- [17] A. Vera and L. Kuntz, "Process-based organization design and hospital efficiency," *Health Care Management Review*, vol. 32, pp. 55–65, 2007.
- [18] P. Harmon, *Business Process Change*. Burlington, MA: Morgan Kaufmann Publishers, 2007.
- [19] H. J. Schmelzer and W. Sesselmann, *Geschäftsprozessmanagement in der Praxis*. München: Carl Hanser, 2006.
- [20] F. X. Frei, R. Kalakota, A. J. Leone, and L. M. Marx., "Process variation as a determinant of bank performance: evidence from the retail banking study," *Management Science*, vol. 45, pp. 1210–1220, 1999.
- [21] A. Gustafsson, L. Nilsson, and M. Johnson, "The role of quality practices in service organizations," *International Journal of Service Industry Management*, vol. 14, pp. 232–244, 2003.
- [22] R. Silvestro and C. Westley, "Challenging the paradigm of the process enterprise: a case-study analysis of BPR implementation," *Omega*, vol. 30, pp. 215–225, 2002.
- [23] S. Hertz, J. K. Johansson and F. de Jager, "Customer-oriented cost cutting: process management at Volvo," *Supply Chain Management: An International Journal*, vol. 6, pp. 128–141, 2001.